



Informality, enforcement and growth

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ARTICLE INFO

JEL codes:

J46

O43

Keywords:

The informal sector

Growth

Enforcement

Taxation

ABSTRACT

This paper develops a two-sector growth model with formal and informal sectors for an economy that cares about redistribution and illustrates its relationship with the enforcement level. The technology gap and labour rigidity explicate the duality. The state can tax the formal sector to subsidise informal income and finance public infrastructure. Alternatively, enforcement, which is costly and corresponds to a variety of discrete components from the security of property rights and integrity of contracts to control of corruptions, can be chosen to favour the formal sector and discourage the informal sector. It is observed that weaker enforcement required to accommodate some degree of informality, which releases tax burden from the formal sector needed for redistribution, can accelerate growth rate. However, sufficiently weaker enforcement dampens the formal sector expansion and growth rate. The growth rate registers an inverted-U shaped relationship against the enforcement level. The optimum enforcement can, however, be higher without formal labour union and subsidisation. This must be higher for welfare maximisation than that of growth rate, especially when the consumer cares about the quality of enforcement.

1. Introduction

A significant share of economic activities dends the enforcement of rules and survives in the informal sector outside the formal setting. The size of this sector seems to have been thriving in not only developing countries but also the developed world (ILO, 2009; Schneider et al., 2011; Kuehn, 2014; Loayza, 2016; Medina and Schneider, 2018; ILO, 2018). Enforcement – by enacting rules and regulations, which encompass a variety of discrete components, from the security of property rights and integrity of contracts to checks on government and control of corruption – is assumed to be an essential ingredient of economic growth (Haggard and Tiede, 2011). However, unless effectively enforced, the mere existence of rules may not have much effect; and the ability of such rules to foster both economic growth and development in the presence of the informal sector was questioned (Desai et al., 2014; Haggard, 2018). Efficient and strict enforcement that represents a clear set of regulations (e.g., the security of property rights and integrity of contracts that underpin trade and investment) would encourage activities in a formal setting, but it discourages activities outside the sector. The growing informality that essentially persists with some degree of weaker

enforcement prompts us to investigate whether the choice of enforcement level has any implication on economic growth. Therefore, the present paper attempts to develop a theoretical relationship between enforcement and economic growth of a society with an informality that cares about redistribution. It seems to be an important exercise when the ability to absorb the entire working population in the formal sector has shrunk worldwide.

The presence of the informal sector has strongly hit the foundations of conventional public policy. The literature no longer tends to discuss public investment only, rather shows concerns more of its efficiency level. In a pioneering work, De Soto (1989) argues that the informal sector thrives on avoiding the cost of tight rules and regulations under formality. Since then, researchers have investigated various issues about the existence of informal sector, and they have supported this conjecture broadly (e.g., Rauch, 1991; Choi and Thum, 2005; Johnson et al., 1997). Similarly, Antunes and Cavalcanti (2007) argue that stronger rules and regulations raise the start-up costs that bind the size of the formal sector. In parallel, many studies suggest that poor institutions with weak rules and regulations limit the incentive to invest in formal settings (Rodrik, 1997; Douglas et al., 2014; Besley, 2017). In a different context, Cooray

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¹ Chandril acknowledges CDE-IEG fellowship during 2017 to support this research. We are extremely grateful to the editor and the two anonymous referees for offering insightful comments on earlier drafts. Usual disclaimers apply.

et al. (2017a) show how among African countries better institutions play a vital role in deriving a favourable impact of openness in work participation. These two sets of observations divulge a dilemma over the public choice between taxation and enforcement in connection with fostering growth and the welfare of society with some degree of informality. In other words, higher taxation and strict enforcement may raise the size of the informal sector, but it needs social security from the state. Increased taxation can finance this. Then, the economy needs to accept a lower level of formality that arises from the distortionary effect of higher taxation. On the other hand, a lower level of taxation encourages employment in the formal sector, but it forces the state to disregard welfare and redistributive measures to some extent. This is a classic example of the efficiency-equity trade-off in the public policy debate, which has received renewed interest in recent years. While the recent global financial and fiscal crises in a couple of countries, including the one that severely affected Greece's economy during 2014–15, recommend a reduction of the fiscal deficit, the rising inequality and informality in most countries during the past two decades suggest the need for redistributive measures. The crises led to limit the fiscal deficit by either raising taxes or cutting expenditure (Alesina et al., 2017). At the same time, growing inequality and informality demand more fiscal resources for redistribution. This inspires us to investigate whether the level of enforcement can be treated as a choice or strategic variable, in addition to taxation, for stimulating growth with better redistribution.

It is evident that the enforcement level varies widely across countries (Antunes and Cavalcanti, 2007; Kuehn, 2014). The standard view suggests that economies with a large informal sector are incapable of implementing rules and regulations effectively because of various structural and socioeconomic bottlenecks and imperfections (Castells and Portes, 1989; Lewis, 1954; Hart, 1973; Amaral and Quintin, 2006; Rodrik, 1997; Acemoglu and Johnson, 2005). Government bureaucracies are often associated with red tape, corruption and lack of incentives, due to asymmetric information in the credit market, which leads to rising misgovernance (Banerjee, 1997). Further, weak enforcement instigates financial friction and, thereby, expands informality (Amaral and Quintin, 2006; Antunes and Cavalcanti, 2007). Reducing enforcement – in the form of regulations such as taxes, entry compliance and red tape – substantially improves welfare and raises consumption (Prado, 2011). Similarly, Ordóñez (2014) accounts for a significant loss of output and productivity in Mexico due to incomplete tax enforcement and allied distortions. Using an endogenous growth model, D'Erasmo and Boedo (2012) predict that countries with a low degree of debt enforcement reveal allocative inefficiency and, hence, show a larger output share produced by low-productive informal-sector firms. In a slightly different context, Cooray et al. (2017b) show how across countries media restrictions limit governance and social outcome.

There is another view, which seems to be a bit unconventional. But, it has received much interest in recent years. Political and strategic motives of a government could be the reason for the informal sector's existence (see, for example, Marcouiller and Young, 1995; Marjit et al., 2006; Dasgupta and Marjit, 2006; Sarkar, 2006). Marcouiller and Young (1995) and Sarkar (2006) argue that a government derives benefits from the existence of the informal sector and hence keeps it alive deliberately. Marcouiller and Young (1995), Marjit (2003) and Choi and Thum (2005) argue that a government with high informality usually allows activities to flourish outside the formal sector as a conscious strategy to tackle poverty and unemployment without putting much burden on taxpayers. Following this logic, Marjit et al. (2006) argue with a simple political support model that a society with lower income inequality and poverty is likely to choose a higher tax rate than a society with greater income inequality and poverty even if both are equally concerned about the social impact of poverty. Maiti and Mukherjee (2013) provide a strategic reason for poor governance that attracts foreign direct investment under trade competition. Hence, the degree of informality could be assumed to depend on either enforcement incapability or strategic motive. Formal investment and employment depend very much on the level of

enforcement irrespective of the reasons for variation. While the existing research has explored the implication of deliberate enforcement choice on poverty, unemployment and other development outcomes, its impact on economic growth has not been adequately discussed.

Enforcement corresponds to a wide and discrete range of regulatory issues. The above discussion reveals that scholars have defined it in various ways to contextualise different situations; hence, they have reached different conclusions. To the best of our understanding, those arguments can be categorised into two broad dimensions. One canvases regulatory dimensions (e.g., taxation, financial and labour market rigidities), which correspond to a certain degree of the burden imposed upon the formal establishment. The other encompasses dimensions that affect the level of security and integrity of economic transactions in the formal setting. In this paper, the enforcement corresponds to regulations that affect the level of security of property rights, the integrity of contracts and checks of corruption that motivate investment and trade in a formal set-up. Better enforcement is similar to what Coase (1937) refers to lower 'transaction costs', and Douglas (1990), North (1991) means better 'institutions'. Weaker the enforcement level, lower the incentive to undertake activities in the formal sector, and greater the stimulation of extra-legal and informal activities. Hence, because enforcement secures investment and reduces transaction costs, it affects formal production favourably. Loayza (1996) also shows theoretically that the informal sector limits the availability of congestible public services for formal firms. Johnson et al. (1997) argue that an index of legal safeguards for investment is inversely related to the share of the unofficial or informal sector. Here, our definition of enforcement does not relate to any form of regulatory rigidities and burdens for formal firms (concerning labour, financial and business regulations). In a typical economy, workers who do not find employment in the formal sector crowd the informal sector, and most of them fall below the poverty line. Such informality is necessarily associated with an unpleasant and disgraceful image of an economy and reveals a degree of social insecurity. Under this circumstance, a government may not prefer to enhance the level of enforcement even if they could. This allows the informal sector to survive to some extent and reduces the need for redistributing resources and, therefore, has implications for economic growth.

To show the relationship between enforcement and growth, we develop a standard two-sector growth model with the formal and informal sectors. Formal sector output is affected negatively by taxation but positively by the level of enforcement. On the other hand, enforcement affects informal income adversely. The state levies taxation on the formal sector to invest in public infrastructure and subsidise informal income. If the enforcement is treated as a choice variable, a relatively weaker level offers a certain degree of regulatory concession to some informal activities for survival, which may discourage formal sector activities. Given these assumptions, the model derives the relationship between taxation and enforcement, and their implication for growth rate. Typically, a labourer in the formal sector, where technology is superior, earns a much higher wage than in the informal sector (ILO, 2009). The wage would be even higher if formal sector workers participate in a union. In the absence of an informal sector, workers outside the formal sector would not find employment, and hence need social security for survival. If the size of this population is substantial, the state cannot ignore them. The state often becomes unpopular by ignoring them and faces a potential threat to its incumbency. Moreover, the state may be unable to handle the unrest. There are ample instances of such a phenomenon (e.g., Kaufmann et al., 2000; Ngo, 2008). In such a situation, a benevolent government should not only optimise growth rate; they must also find a redistributive strategy that improves the aggregate social welfare when the conventional redistributive measure is not successful. If strict enforcement does not allow economic activity in the informal sector, taxation must be high enough to subsidise the income of informal sector workers, at least for those who fall below the poverty line. If not, the state could be unpopular, which it does not want.

Even if the state is capable, simple logic suggests that in the presence

of a large informal sector, a government that cares about redistribution must levy higher taxes to deal with high inequality and poverty. To the best of our knowledge, no empirical study confirms this claim. Rather, it has been observed that taxation is higher in more egalitarian societies where informality is lower (like in Europe) than in more unequal societies where informality is higher, like in Brazil (Persson and Tabellini, 2000; Drazen, 2000). Hence, the state has to choose what and how much. A rich and interesting set of frameworks is available in the literature, which theorises diverse issues of regulations and enforcement; two of them are overlapping generation model (Amaral and Quintin, 2006) and matching model (Antunes and Cavalcanti, 2007). Strictly speaking, none of them considers the dimension of enforcement that corresponds to a level of property rights, security, the integrity of contracts and control of corruption as a strategic variable in a growth model. Similar to Johnson et al. (1997), we assume two sectors, formal and informal. The formal sector could be either taxed more to subsidise outside workers or given a tax rebate if outside workers are allowed to survive to some extent. Moreover, formal sector labourers join a union in compliance with legislative support and, hence, earn rent. This creates an additional distortion on top of technological duality and is responsible for higher unemployment in this setting. If the state does not offer social security, they may remain unemployed. The model captures the allocation of labour between two sectors through the bargaining process of formal workers, which is essentially affected by the choice of taxation and enforcement levels. Together, they ultimately determine the employment composition of the economy and resultant growth.

Using such a framework, we find that taxation and enforcement are positively related. If the level of enforcement is raised, the informal sector suffers a loss, and taxation needs to be increased to subsidise the loss of the informal sector. However, too weak enforcement can damage formal sector production more than the benefits from a tax reduction. Hence, the growth rate shows an inverted-U shape against the level of enforcement. The growth-maximising enforcement level falls between the two extreme levels. As enforcement rises, two opposite forces work on growth. One, output grows fast in the formal sector. This is the productive sector of the economy. On the other hand, informal production declines with the rise in enforcement, and the increased informal sector demands higher taxation from the formal sector to compensate for the loss of their income. Taxation needs to be even higher in the presence of a formal sector labour union because of formal sector workers bargain for a higher wage with the increased union bargaining power. In effect, higher taxation along with increased union wages crowd out formal sector production and, hence, limit growth. As long as the former effect is stronger, the growth rate accelerates. However, the favourable effect of enforcement tends to be weakened with higher enforcement, and the distortionary effects of taxation and bargaining power gradually become stronger. Hence, there could be a unique level of enforcement within the ranges that can maximise growth. If the level of enforcement is weakened a bit to allow some economic activities in the informal sector, it eases away from the tax burden on the formal sector, improves redistribution and helps the economy to grow. However, reducing the enforcement level too much may severely hurt formal sector expansion and limit growth rate. In case the government does not want to subsidise informal sector income, the optimum enforcement for growth becomes independent of bargaining power and, hence, must be higher than the one with subsidy and labour union.

The optimum enforcement level that maximises welfare is found to be greater than the one that maximises growth because enforcement affects informal income and, thereby, the initial level of consumption that enters into welfare estimation. If individuals prefer better enforcement, the optimum level would be even higher. These results suggest that deliberately weak enforcement may raise growth rate and welfare in an economy with a large informal sector, but both could be dampened if enforcement is too weak. Using standard parameters, we calibrated growth rate and welfare functions; they show an inverted-U shaped relationship against the level of enforcement.

The rest of the paper is organised as follows. The next section provides some facts and figures on informality, enforcement and growth rate. Section 3 describes the formal model. Section 4 simulates the main relationship using standard parameters available in the existing literature. The last section ends with concluding observations.

2. Facts

Though it is very hard to measure the informal sector precisely, the existing research still tried it, and often they are not strictly comparable. In general, the share of the informal economy in developing countries is much higher than in developed countries. In a typical developing economy, the informal sector produces about 35% of the gross domestic product (GDP) and employs 70% of the labour force (Loayza, 2016). As per the latest figures, ILO (2018) estimates that 85.8% of employment is informal in Africa. The proportion is 68.2% in Asia and the Pacific, 68.6% in the Arab States, 40.0% in the Americas and 25.1% in Europe and Central Asia. The report also reveals that 93% of the world's informal employment is in emerging and developing countries. Earlier, Schneider and Enste (2000) estimated the size of the shadow economy (as a percentage of GDP) for various countries in the early 1990s. They showed it varies hugely across countries – 76% in Africa, 60% in Central and South America, 70% in Asia, 28% in Central Europe, 43% in the former Soviet Union countries and 30% in OECD countries. Jütting and Laiglesia (2009) also concluded that more than half of all jobs in the non-agricultural sectors of developing countries could be considered informal. Later, Schneider (2012) calculated the size of this sector as a percentage of official GDP for 21 OECD countries and found that it varies from 7.2% in the USA to 25.1% in Greece with an unweighted average of 13.9% for 21 OECD countries. According to Kuehn (2014), approximately 16.7% of output in high-income OECD countries is too produced informally. Using a structural hybrid method, Medina and Schneider (2018) estimated a new series of shadow or informal economy for 158 countries over the 1991–2015 period. The average size of the shadow economy is 31.9% of GDP; the largest ones are Zimbabwe (60.6%) and Bolivia (62.3%), and the lowest ones are Austria (8.9%) and Switzerland (7.2%).

It is difficult to draw a conclusive statement from its pace over time because comparability is limited. National governments and scholars have used different methodologies to estimate the size of the shadow economy. Schneider et al. (2011) estimated the size of the shadow economy (as a percentage of GDP) across countries for the 1999–2006 period; in a group of 88 developing countries, average size increased from 35.7% in 1999 to 37% in 2006. Using comparable data available for the 2004–2016 period, ILO (2018) estimated the size of the informal economy for some countries, but this does not convey any conclusive trend. While the share of informal employment has declined in some Latin American countries like Argentina and Peru, it has increased in Europe, Central Asia, Russia and Serbia.

Although the informal sector was viewed to be in dire straits with the economy's progress, Baily et al. (1998) point out that labour-intensive traditional forms of production, including those in the countryside, are not disappearing, but persisting as market-led industrialisation is intensifying in developing countries. However, Loayza (1996) empirically shows the negative effect of the informal sector on economic growth in Latin American countries. Eilat and Zinnes (2000) further show that a decline in the informal sector accompanies the growth of GDP. On the other hand, Sarte (2000) theoretically argues that the informal sector itself is not detrimental to growth; rather, bureaucratic rent-seeking behaviour, in the presence of the high extra-legal cost associated under the environment of fragile property rights, lowers growth. Moreover, using an endogenous model, Nabi and Drine (2009) show that under certain conditions the informal sector can expand with growth. Using a large panel data set of 161 countries over the 1950–2010 period, Elgin and Birinci (2016) obtain an inverted-U relationship between the size of the informal sector and economic growth. Schneider (2006) also shows

that the shadow economy negatively and significantly affects growth in developing countries but positively in developed OECD and transition countries. When plotted against informal employment (a proxy of vulnerable employment share), five-year average growth figures do not convey any relationship for the 1990–2009 period but tend to show an inverted-U shape² during the 2010–17 period (see Fig. 1a and b). The plotted lines confirm that growth does not seem to have fallen with the rise of informal employment, at least up to a certain level.

Such a relationship between informality and growth rate persuades us to look at the connection between enforcement and taxation. Taxation (as a percentage of GDP) displays a positive relation with two proxies of enforcement level (i.e., governance effectiveness and regulation quality) in Fig. 2. Data on taxation, governance and other variables have been drawn from World Governance Indicators and World Development Indicators. Across countries, the level of enforcement represents the quality of regulations and the effective implementation of the legal system and regulations. Higher the value, greater the degree of legal security and contractual integrity; lower the value, weaker the enforcement level. These indicators show a positive relationship with taxation (as a percentage of GDP), suggesting that higher taxation accompanies higher enforcement.

Moreover, the growth rates are plotted against enforcement indicators. This does not display a monotonic and positive relationship. Rather, an inverted-U shaped growth seems to be pronounced especially during the 2010–2017 period (see Fig. 3). This motivates us to investigate why growth rate may tend to be lower around the highest level of enforcement.

Since a simple correlation cannot confirm the relationship, regression results have been presented. To find out the effect of enforcement and taxation on growth rate in the presence of the informal sector, one should undertake an econometric exercise and control some relevant factors. Using the instrumental variable regressions, enforcement and taxation are regressed separately along with their interaction terms with informality on log GDP (see Table 1). These apart, government consumption share (as a percentage of total consumption), index of the sound monetary system (combining average inflation, money growth and fluctuations of inflation) and trade openness (combining average tariff rate, hidden trade barriers, costs of trading) are controlled in all the regressions. Note that the coefficients of enforcement and its interaction term with informality are statistically significant. They further indicate positive and negative relation, respectively. The results suggest that relatively lower enforcement in the presence of informality raises growth.

Further, the coefficients of three interaction terms between enforcement, taxation and informality are found significant and positive, meaning that the enforcement level starts raising growth along with increased taxation. Since higher taxation helps redistribution effectively, enforcement must rise, but the extent of informality limits it. This result is true in all three types of panel regressions (see Table 1).

3. The model

We model a typical economy, where a single final good is produced in both formal and informal sectors simultaneously with different production technologies. Technology in the formal sector is better than that used in the informal sector. It is evident that the informal economy is more likely to be populated by low-skilled individuals. There are three economic agents in the model – households, government and labour unions. Labourers prefer higher-wage and formal-sector jobs. A similar assumption is made in Carruth and Oswald (1981) and Marjit (2003) in a dual

economy model with the union and non-union workers. For simplicity, we assume homogeneous labour, which is a better representation of reality. The heterogeneity assumption in the labour force will complicate the mathematical analysis but not add anything in this framework, as the model does not focus on skill. So, the model proceeds with the assumption of homogeneous labour, which implies that getting a job in the formal sector is a matter of luck. Since there is no difference between workers, they are selected in the formal sector randomly from the unemployed pool. The details of each economic actor are described below.

3.1. Formal sector

The formal sector consists of a fixed number of competitive firms, which produce final goods using private capital (K), labour (L) and non-rival and non-excludable productive public goods, G (mainly infrastructure). The government can affect the formal sector output, Y_F , by choosing a level of enforcement, g , which represents the enforcement level that corresponds to a variety of discrete components from the security of property rights, in the integrity of contracts to control corruptions. The existing literature talks about various dimensions and their effect on production (Rodrik, 1997; Choi and Thum, 2005). Due to its multidimensional nature, scholars define the enforcement differently as per the requirement of their models. For example, Amaral and Quintin (2006) consider the enforcement level in relation to financial contracts. In their model, there is a contract enforcement gap between the formal and informal sectors, and it is more difficult to enforce contracts in the informal sector. Under various other situations, like tax evasion, debt and financial friction, similar frameworks are available in the literature (e.g., Antunes and Cavalcanti, 2007; Bosch and Esteban-Pretel, 2012; Kuehn, 2014). Jonasson (2011) further used the term ‘governance’ to represent the extent to which the government supports the market transactions. Our formulation is similar to what Marjit et al. (2006) specify. Relatively weak enforcement helps informal agents, but hurts people with tax-paying capacity, whereas strong enforcement protects taxpayers and formal transactions legally and effectively. In this paper, the term ‘enforcement’ is defined as a set of regulatory actions affecting activities in both formal and informal sectors, but differently. If the enforcement is higher, the government can limit the size of informality and vice versa.

The production function of the formal sector is given by

$$Y_F = AK^\alpha L^\beta G^{1-\alpha} \mu(g) \quad (1)$$

where $\alpha, \beta, \alpha + \beta \in (0, 1)$; $\mu'(g) > 0$, $\mu''(g) < 0$.

Following Barro (1990) and many of its extensions, G is assumed as a flow variable, for simplicity. In reality, it is a stock variable. The enforcement, g , is treated as a flow variable. The formal output rises with g , but at a decreasing rate. The evidence from the World Bank study supports this assumption (Acemoglu and Robinson, 2008). Here, A denotes the level of technology and ignores technological progress. In the model, α and β represent the output elasticities concerning capital and labour. Following Chang et al. (2007), the Cobb-Douglas production function is assumed to satisfy the increasing returns to scale for all inputs but at decreasing return from private inputs. This implies that supernormal profit exists after paying factor payments according to their marginal productivities. A positive profit is treated as a rent to be negotiated between union and firm in this sector. Higher the bargaining power of workers, lower the profit. To keep the number of firms constant, we assume, following Chang et al. (2007), that there exists a fixed factor (e.g., land). For the sake of simplicity, it is normalised to unity.

The formal representative firm maximises its profit, π , defined as

$$\pi = Y_F - w_F L - rK \quad (2)$$

Here w_F and r stand for the wage of formal labour and the rental rate of private capital respectively. As there is a single final good, its price is normalised to unity.

² The fitted line provides locally weighted scatterplot smoothing values, which are obtained by running a regression. This regression is undertaken so that the central point gets the highest weight and points that are farther away receive less weight (Cox, 2005).

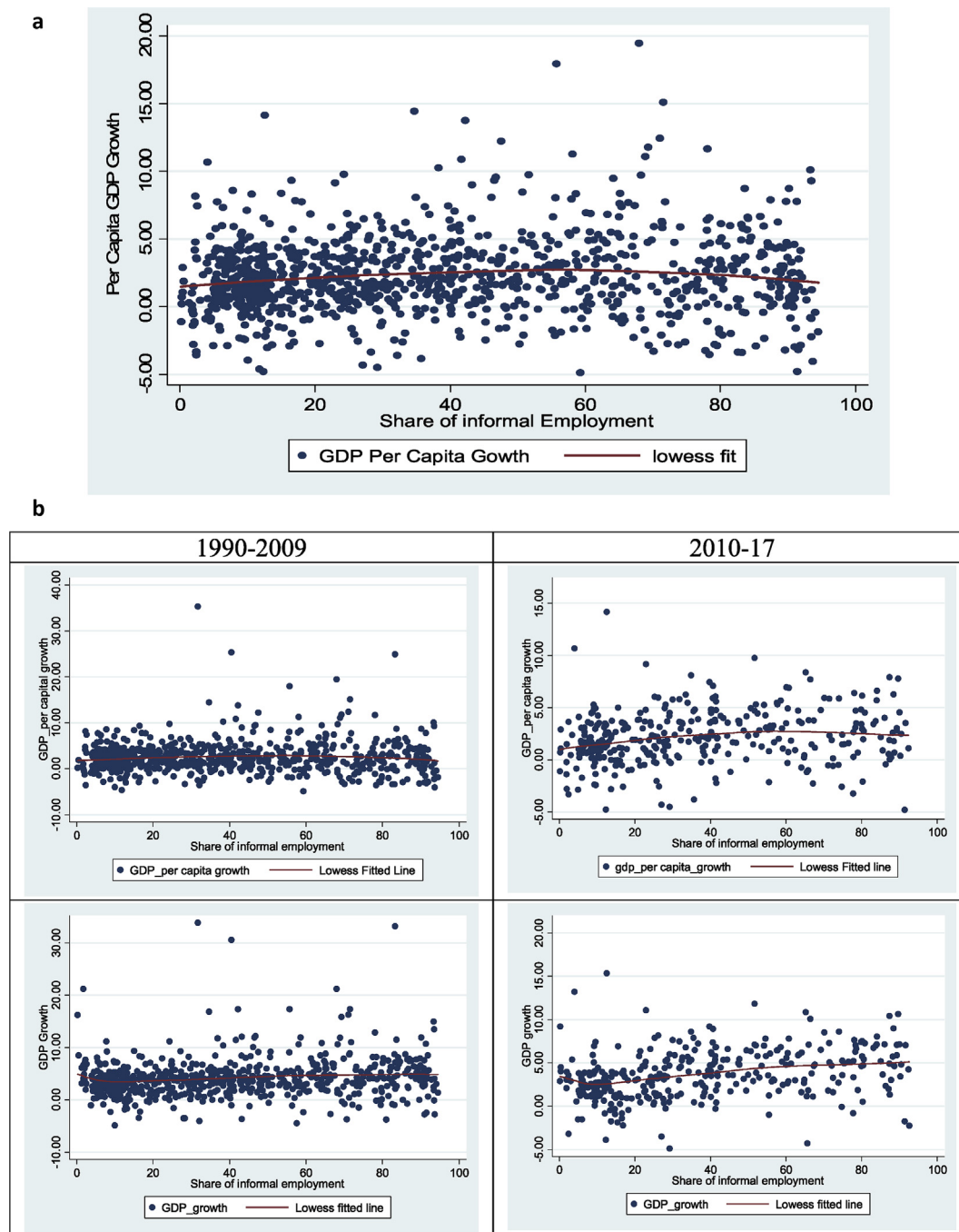


Fig. 1. a: Relationship between the share of informal employment and growth (1990–2017). b: Relationship between the share of informal employment and growth rate.

Source: Authors' calculation (WDI and WGI, World Bank data)

3.2. Informal sector

The unlucky workers who cannot find a job in the formal sector join the informal sector. They could be engaged in extra-legal activities for survival, but not in criminal or illegal works. We assume that informal production does not need private capital. This is an approximation of reality, in the sense that the informal sector requires much less capital than the formal sector. Also, informal sector firms are usually not financed by banks and do not accumulate capital assets. This is evident from a survey in 1983 of 10,000 households in Lima, discussed by [Thomas \(1992\)](#). The survey reveals that almost 50% of informal workers

operate with less than US\$500 of capital per head, whereas 90% of a comparable sample of formally employed workers operate with more than US\$6000 of capital per head. [Söderbaum and Teal \(2000\)](#) collected survey data from Cameroon, Ghana, Kenya and Zimbabwe and pointed towards this difference. The study suggests that firms that employ more than 100 workers operate on average at three to four times higher physical capital per worker than firms that employ fewer than six workers. [Amaral and Quintin \(2006\)](#) provided a theoretical explanation for why formal firms operate at a higher physical capital-to-labour ratio than informal firms. The report on the unorganised sector of the Indian economy suggests that more than 95% of enterprises in the informal

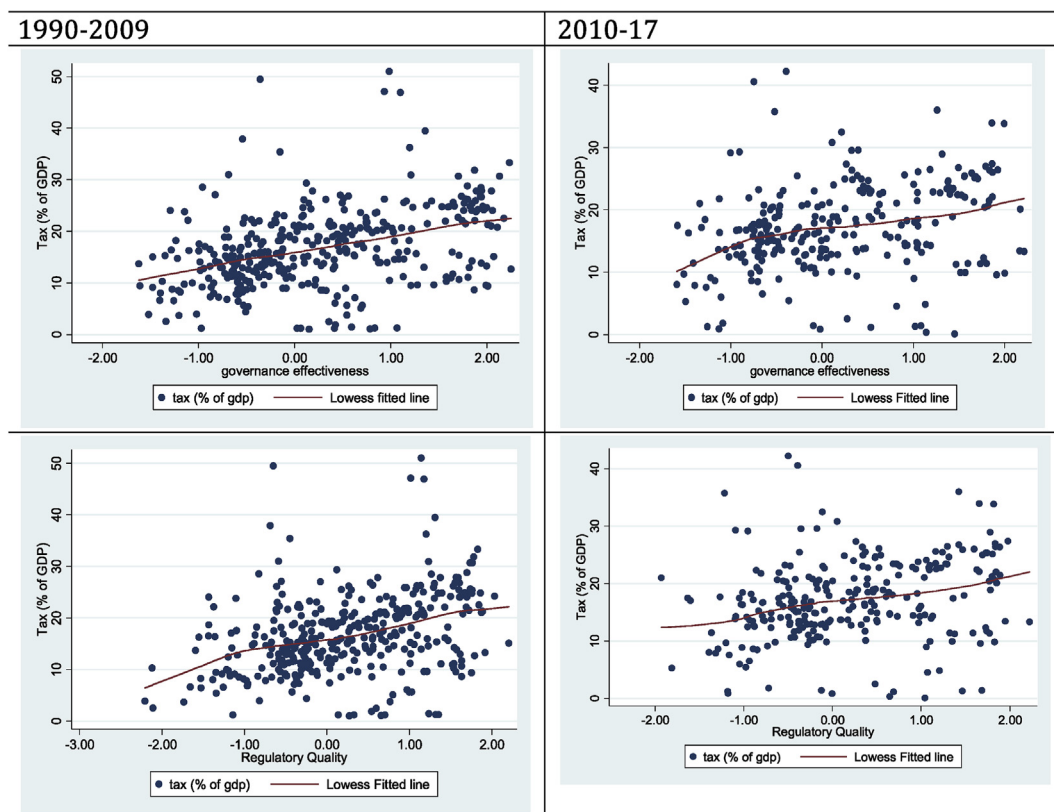


Fig. 2. Taxation and enforcement across countries.

Source: Authors' calculation (WDI and WGI, World Bank data)

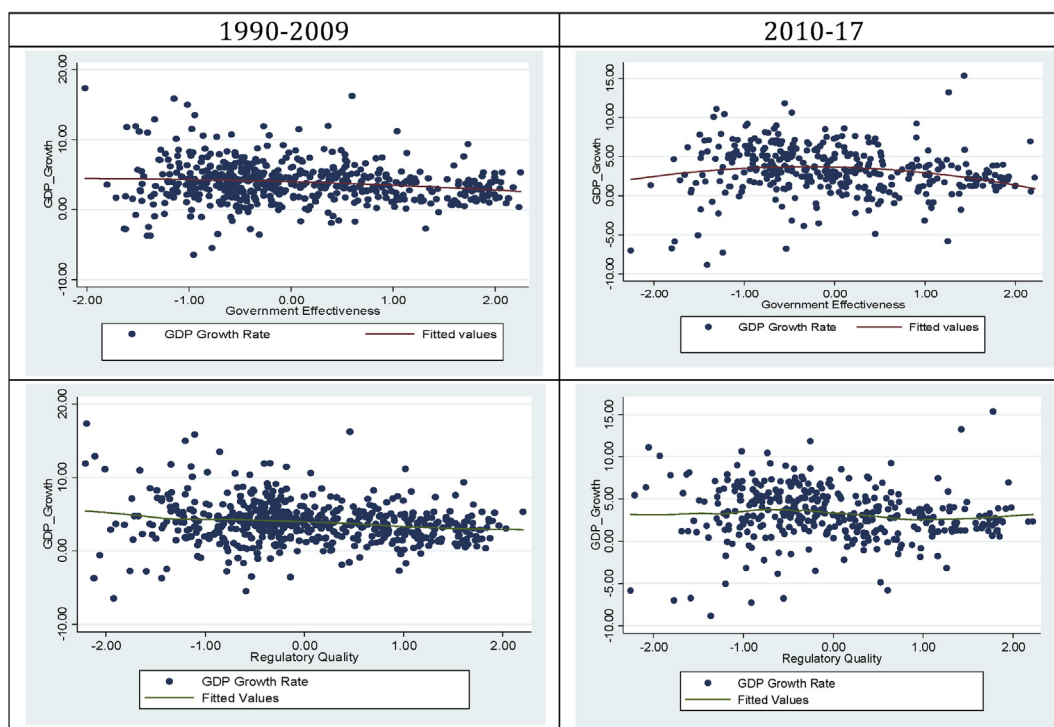


Fig. 3. The level of enforcement and GDP growth rate across countries.

Source: Authors' calculation (WDI and WGI, World Bank data)

Table 1

Enforcement and taxation on growth (lnGDP) in the presence of informality.

Variables	2SLS	LIML	GMM
Enforcement	0.24***	0.24***	0.23***
Enforcement*informal	−0.003***	−0.003***	−0.003***
Tax	0.018	0.018	0.02
Tax * informal	−0.001***	−0.0007***	−0.0007***
Enforcement * tax * informal	1.95×10^{06} ***	1.95×10^{06} ***	1.7310^{06} ***
Govt. cons. share (%)	0.01**	0.02**	0.02**
Sound monetary system	0.16***	0.16***	0.15***
Trade openness	0.19***	0.19***	0.17***
Constant	7.4***	7.39***	7.27***
Instrumented variables	Enforce, tax	Enforce, tax	Enforce, tax
Instruments	Lag (enforce, tax and regulations)	Lag (enforce, tax and regulations)	Lag (enforce, tax and regulations)
Obs.	248	248	248
R ²	0.84	0.84	0.84
Wald chi	1809	1809	1907

Note: *p < 0.10, **p < 0.05, ***p < 0.01.

sector did not receive formal finance during 2005–06 (NSSO, 2005).

We also assume that the output of the informal sector, Y_I , varies proportionally to the number of informal workers³ and public expenditure, G . Jonasson (2011) empirically shows the positive impact of public investment on the informal sector. We assume that the total labour force is equal to unity and that it does not grow over time. So, $(1 - L)$ is the number of workers absorbed in the informal sector. The production function of this sector is given by

$$Y_I = B(1 - L)G\gamma(g); \gamma'(g) < 0, \gamma''(g) < 0 \quad (3)$$

We already assume that, technologically, the informal sector is backward compared to the formal sector. The technology parameter, B , is also time-independent. Similar to Marjit et al. (2006), we assume that informal output falls as the level of enforcement rises, but faster. If the enforcement (i.e. g) is high, informal firms bear higher costs, due to extra-legal costs (e.g. bribing) to avoid rules. On the other hand, if the government keeps g low (e.g. allowing congestion of vendors in streets, tolerating illegal electricity connections and allowing slum activities and running informal activities in public spaces), it helps the informal sector to flourish. In reality, there exist informal activities that are not affected by the level of enforcement. Our model considers only those that are affected inversely by the enforcement level. Like Marjit et al. (2006), this paper assumes that their substitutability dominates the complementarity between the formal and informal sectors. Loayza (1996) also empirically shows that government strength and efficiency significantly and negatively influence the informal sector. Jonasson (2011) uses data on Brazil to show that government effectiveness negatively affects the informal sector. Since there is only one private input in the informal sector, the whole output is paid to informal workers. So, the informal wage, w_I , is given by

$$w_I = BG\gamma(g) \quad (4)$$

Note that informal wage is directly affected by technology and investment on public infrastructure, and negatively influenced by the enforcement level.

3.3. Private capital market

The rental rate of private capital is determined by the equality between demand and supply in a competitive environment. Profit maximisation of the competitive firm finds the following demand function for capital.

³ Mazumder (2015) also assumes that output in the unorganised sector varies proportionally with labour input.

$$r = A\alpha K^{\alpha-1} L^\beta G^{1-\alpha} \mu(g) = \frac{\alpha Y_F}{K} \quad (5)$$

Similar to the standard literature, the rental income is equal to the capital share of the formal sector.

3.4. Government

The government taxes wage, rental and profit incomes generated in the formal sector. This tax revenue is spent on procuring public goods and services and subsidising informal sector workers. The government maintains a balanced budget in each period and spends a fixed fraction of formal income, $\bar{\tau}_G$, to finance public goods and services, for simplicity. We assume, further, that it is exogenously given. It is shown in the equation below.

$$\bar{\tau}_G Y_F = G \quad (6)$$

Following Marjit et al. (2006), we assume that enforcement is costly and that the cost increases with the level. We denote the cost per unit of G by $Z(g)$, where $Z'(g) > 0$ and $Z''(g) > 0$.⁴ The subsidy given to informal workers may not be a direct monetary transfer; it could be a subsidy on food, fuels, healthcare services, education or various concessions for public services. So, the government's budget constraint becomes

$$\tau Y_F = G[1 + z(g)] + S(1 - L) \quad (7)$$

Where S is subsidy per informal worker, and τ is the proportional income tax rate. We assume that S is proportional to formal sector output,⁵ i.e., $S = sY_F$. This suggests total subsidy rises with formal production even when the subsidy rate is fixed. So, the government budget constraint per unit of formal output is given by

$$\tau = \bar{\tau}_G[1 + z(g)] + s(1 - L) \quad (8)$$

Note that taxation depends on how much is distributed between public infrastructure, cost of enforcement and subsidy given to informal sector workers.

3.5. Labour union and wage bargaining

We assume that labour union exists only in the formal sector. In addition to the technological difference, this is another source of imperfection/distortion in the model. The government undertakes redistributive measures to minimise the cost of such distortions. There are other kinds of labour market imperfections. We model the labour union. However, the results derived in this model would not be very different from those types of labour market imperfections. For example, an increase in wage reduces formal employment under efficiency wage and minimum wage laws as well. We assume a closed shop labour union here. This implies that only members of a union are eligible for getting employment in the formal sector. As a result, firms cannot hire workers from outside the union. This enables the union to exercise its bargaining power to raise wages over the competitive one. Under the efficiency wage, formal firms themselves do not hire workers at a lower wage to

⁴ Here, the cost of enforcement is linked to public goods spending. Wage increases at a rate the formal output rises in the steady state. Government spending is assumed to be a fixed fraction of formal output. So, the ratio of wage relative to public spending is time-independent. As their wages are proportional to public spending, the cost of enforcement can be expressed as a proportion of the public goods, and the cost of providing the same level of enforcement rises with time.

⁵ For simplicity, we assume that the subsidy per informal worker is a constant fraction of formal output. This subsidy rate is a choice variable of the government. In subsection 4.4, we consider a case of variable subsidy rate. Moreover, growth equation (21) shows that the growth rate always varies inversely with the subsidy rate.

avoid efficiency loss, and the wage gap between the two sectors prevails. To model the behaviour of a labour union, we follow Chang et al. (2007) and Bhattacharyya and Gupta (2015, 2016), and consider a Stone–Geary form of the union utility function. We further assume that both employment and wage gap after tax, arising from the difference between formal and informal sector earnings, are its arguments. The utility function of the labour union is given by

$$u_T = [(1 - \tau)w_F - w_I]^m L^n \quad \text{with } m, n > 0. \quad (9)$$

Here, u_T denotes the utility of the labour union. Moreover, m and n are two non-negative preference parameters representing elasticities of union utility with respect to wage and employment respectively. If $m > (<) (=) n$, then the labour union is said to be ‘wage-oriented’ (‘employment-oriented’) (‘neutral’).⁶

Here, the ‘right to manage’ model of bargaining is considered. It is more common in the real world.⁷ In this model, the union and firm bargain over the formal wage. Then, the employment is solely determined by the firm from its labour demand function. The result can be obtained by maximising the ‘generalised Nash product’ function, which is given by

$$\psi = (u_T - \bar{u}_T)^\theta (\pi - \bar{\pi})^{(1-\theta)} \quad \text{with } 0 < \theta < 1. \quad (10)$$

Here, $\bar{u}_T$ and $\bar{\pi}$ represent, respectively, the fall-back utility of the labour union and the profit of the firm. Further, we assume, for simplicity, that bargaining disagreement stops production. This gives no profit and utility at zero employment. The parameter θ represents the relative bargaining power of the union. An increase in θ can also be viewed as a rise in labour market outcomes.

The formal labour demand function can be obtained by maximising the profit, given by equation (2), concerning L ;

$$w_F = \beta AK^\alpha L^{\beta-1} G^{1-\alpha} \mu(g). \quad (11)$$

So, incorporating Eqs. (1), (2), (9) and (11) into Eq. (10), we obtain

$$\psi = \{[(1 - \tau)\beta AK^\alpha L^{\beta-1} G^{1-\alpha} \mu(g) - w_I]^m L^n\}^\theta \{(1 - \beta)AK^\alpha L^\beta G^{1-\alpha} \mu(g) - rK\}^{(1-\theta)} \quad (12)$$

Equation (11) shows a monotonically inverse relationship between formal wage and employment. It is, therefore, equivalent to the maximisation of Eq. (12) concerning L . Now, using the first-order condition for maximisation of Eq. (12) along with Eqs. (1), (4)–(6) and (8), we obtain the negotiated level of formal employment, L^* , given by

$$L^* = \frac{\{1 - \bar{\tau}_G[1 + z(g)] - s\}^\Theta}{\{\bar{\tau}_G B\gamma(g) - s\Theta\}} \quad (13)$$

Here, Θ is given by

$$\Theta = \frac{\beta\{\theta n(1 - \alpha - \beta) + \beta(1 - \theta)(1 - \beta) - \theta m(1 - \beta)(1 - \alpha - \beta)\}}{\{\theta n(1 - \alpha - \beta) + \beta(1 - \theta)(1 - \beta)\}}. \quad (14)$$

No strict restriction over the preference of the labour union is imposed, but it should not be excessively wage-oriented. Or, m is very much higher than n . If the union opts for a very high wage, it solves for lower formal employment. So, the numerator of Θ is positive, which implies that Θ is a positive fraction (see Appendix A). A rise in the relative bargaining power of union lowers Θ as shown below.

$$\frac{\partial \Theta}{\partial \theta} = - \frac{\beta^2(1 - \beta)^2 m(1 - \alpha - \beta)}{\{\theta n(1 - \alpha - \beta) + \beta(1 - \theta)(1 - \beta)\}^2} < 0. \quad (14a)$$

Since the total labour force is equal to unity, the negotiated level of formal employment, L^* , is assumed to be a positive fraction. To find $0 < L^* < 1$, we need that both the numerator and denominator of Eq. (13) are to be positive, and that the denominator is to be greater than the numerator. So, these conditions are assumed to be satisfied (see Appendix B). The conditions impose an upper bound on the level of enforcement for given $\bar{\tau}_G$ and s . This suggests that the enforcement level raises taxation and can rise so much that formal output could be extremely low. These extreme situations are restricted by these assumptions. Throughout the paper, the range of g always is assumed to be satisfied under all such conditions. Given these conditions, Eq. (13) shows that the share of a negotiated level of formal employment is time-independent; so is the share of informal employment. From Eq. (13), we also obtain

$$\frac{\partial L^*}{\partial s} = - \frac{\{\bar{\tau}_G B\gamma(g) - \Theta + \Theta \bar{\tau}_G[1 + z(g)]\}^\Theta}{\{\bar{\tau}_G B\gamma(g) - s\Theta\}^2} < 0 \quad (13a)$$

$$\frac{\partial L^*}{\partial \bar{\tau}_G} = - \frac{[1 + z(g)]\Theta\{\bar{\tau}_G B\gamma(g) - s\Theta\} + B\gamma(g)\{1 - \bar{\tau}_G[1 + z(g)] - s\}^\Theta}{\{\bar{\tau}_G B\gamma(g) - s\Theta\}^2} < 0 \quad (13b)$$

$$\frac{\partial L^*}{\partial \theta} = \frac{\{1 - \bar{\tau}_G[1 + z(g)] - s\}^\Theta \bar{\tau}_G B\gamma(g)}{\{\bar{\tau}_G B\gamma(g) - s\Theta\}^2} \frac{\partial \Theta}{\partial \theta} < 0 \quad (13c)$$

Eq. (13a) and (13b) show that formal employment varies inversely with subsidy rate and share of public expenditure. Given the level of enforcement, an increase in subsidy demands higher taxation; this limits formal production and employment. On the other hand, a rise in the share of public expenditure demands higher taxation. But, it has two forces. One, higher taxation negatively affects formal production and employment. Second, public expenditure encourages both sectors favourably.

Moreover, public expenditure raises informal sector productivity and, thereby, wages. As a result, unions bargain for higher formal wages in response to a rise in informal wages. This adds to the negative force of formal employment. Hence, the formal employment declines.

Eq. (13c) shows that a rise in the union's bargaining power raises wages and, hence, reduces formal employment. So, the rise in bargained wage leads to a fall in employment. Now, substituting the value of negotiated employment from Eq. (13) in (11), the negotiated wage is obtained. Further, using Eqs. (1), (4), (6), (8), (11) and (13), it can be shown that

$$(1 - \tau)w_F = w_I \left\{ \frac{\beta}{\Theta} \right\} \quad (15)$$

Eq. (15) reveals that the formal wage after tax varies positively with informal wage and inversely with Θ or positively with θ . If the labour union is assumed powerless (i.e., $\theta = 0$), then $\Theta = \beta$ or this becomes labour share. Eq. (14a) suggests that $(1 - \tau)w_F > w_I$ when $\theta > 0$. The important result obtained is summarised in the following proposition.

Proposition 1. *Formal employment is inversely related to the informal sector subsidy rate, public expenditure share of GDP and union bargaining power of formal labour.*

Note that Eq. (15) can be written as

$$\frac{(1 - \tau)w_F}{w_I} = \left\{ \frac{\beta}{\Theta} \right\} \quad (15a)$$

⁶ See Chang et al. (2007) to know more about these parameters.

⁷ Oswald (1993) and Oswald and Turnbull (1985) reveal that, in both the USA and Britain, labour unions normally do not negotiate over the total number of jobs.

The net wage gap between formal and informal sectors is independent of policy instruments (e.g. subsidy rate and enforcement level). It is solely determined by the labour contribution driven by the technology used in the formal sector, the relative bargaining power of the union and its preference for wage or employment orientation.

Lemma 1. *The state cannot influence the net wage gap between formal and informal sectors directly by employing any policy instrument as it is solely determined by technology, union bargaining power and its preferences in the formal sector.*

3.6. The household

A representative household derives instantaneous utility from consumption, c , of the final good. We ignore leisure as a source of utility. Total labour supply remains constant. Following Barro (1990) and many of its extensions, productive public goods and services do not enter into the utility function. The household chooses the time path of consumption to maximise her discounted present value of instantaneous utility subject to her intertemporal budget constraint. The household's problem is given below.

$$\text{Max} \int_0^{\infty} \log(c) e^{-\rho t} dt \quad (16)$$

subject to,

$$\dot{K} = (1 - \tau)[w_F L + rK + \pi] + w_I(1 - L) + S(1 - L) - c \quad (17)$$

and $K(0) = K_0$.

Here, c is the control variable, K is the state variable, and ρ is the constant discount factor. Capital is irreversible and does not depreciate over time. The share of formal and informal sector workers remain the same for all households⁸, and the representative household saves the rest of net income over consumption, which contributes to the process of capital accumulation. Household income consists of net return from the formal sector, informal wage and subsidy given to informal labourers. Since all factors of production belong to the household, they all go there.

Solving this dynamic optimisation problem, the balanced steady-state growth rate of consumption, denoted by v , is given below:

$$v = \frac{\dot{c}}{c} = (1 - \tau)r - \rho \quad (18)$$

Eq. (18) shows that the rate of consumption growth is equal to the net rate of return from private capital over the tax and discount rate.

4. The steady-state

The economy always remains at the steady state balanced growth path without any transitional dynamics. In the steady state, L^* , $1 - L^*$, g , r , $\bar{\tau}_G$, τ , s , v , G/K are time-independent, whereas Y_F , Y_I , w_I , w_F , c , G , K , π and S grow at the rate v .

4.1. Optimum enforcement for growth

Let us focus on the growth-maximising level of enforcement at the steady-state growth path. The government objective is to maximise welfare. However, to avoid analytical complexity, we do not derive it. Rather, we first derive the optimum enforcement for growth and later analyse whether it satisfies the welfare maximisation as well. Let us look at the effect of enforcement on formal employment. For that, we use Eq.

(13) and obtain

$$\frac{\partial L^*}{\partial g} = - \frac{\bar{\tau}_G \Theta [z'(g) \{ \bar{\tau}_G B \gamma(g) - s \Theta \} + B \gamma'(g) \{ 1 - \bar{\tau}_G [1 + z(g)] - s \}]}{\{ \bar{\tau}_G B \gamma(g) - s \Theta \}^2} \quad (19)$$

The numerator of the above equation is ambiguous as $z'(g) > 0 > \gamma'(g)$. The following condition is assumed to be satisfied.

Condition 5: $z'(g) \{ \bar{\tau}_G B \gamma(g) - s \Theta \} + B \gamma'(g) \{ 1 - \bar{\tau}_G [1 + z(g)] - s \} > 0$.

Condition 5 depends on the functional forms of $z(g)$ and $\gamma(g)$. Since enforcement is costly and it discourages informal production, this condition suggests that the loss of informal income for marginal improvement from enforcement should not exceed the cost incurred for it. This is a sufficient condition for choosing a level of enforcement. Under this condition, the range of g can be found, which is of our interest. This implies that the enforcement has a negative effect on formal employment. The intuition behind this result is as follows. The labour union derives utility from the formal and informal wage gap. A rise in g raises taxation, because effective enforcement is costly. This lowers the gap. However, higher the level of enforcement, lower the informal income, and widen the wage gap. If the tax effect dominates the enforcement effect, the wage gap decreases with g . As a result, unions bargain for higher wages, leading to a drop in employment.

To obtain the effect of g on taxation, we use Eq. (8) and get

$$\frac{\partial \tau}{\partial g} = \bar{\tau}_G z'(g) - s \frac{\partial L^*}{\partial g} > 0. \quad (20)$$

Eq. (20) shows that enforcement raises taxation. As g rises, the cost of financing enforcement goes up, and it directly demands higher taxation. On the other hand, a rise in g reduces formal employment as the union bargains for higher wages. Higher union wage and taxation raise the cost of formal production further, and thereby reduce formal employment in the aggregate, leading to a rise in informal employment. The rise of informal employment demands higher subsidy, which essentially raises taxation to finance them further.

Now, using Eqs. (1), (5), (6), (8) and (18), the growth equation of the economy in terms of enforcement, g is given by:

$$v = \left\{ 1 - \bar{\tau}_G [1 + z(g)] - s(1 - L^*) \right\} \alpha \left(\bar{\tau}_G \right)^{\frac{1-\alpha}{\alpha}} A^{\frac{1}{\alpha}} (L^*)^{\frac{\alpha}{\alpha}} \mu(g)^{\frac{1}{\alpha}} - \rho. \quad (21)$$

From Eq. (21), we obtain

$$\frac{\frac{\partial v}{\partial g}}{v + \rho} = \frac{-\bar{\tau}_G z'(g) + s \frac{\partial L^*}{\partial g}}{\left\{ 1 - \bar{\tau}_G [1 + z(g)] - s(1 - L^*) \right\}} + \frac{\beta \frac{\partial L^*}{\partial g}}{\alpha L^*} + \frac{1}{\alpha} \frac{\mu'(g)}{\mu(g)} \quad (22)$$

The growth-maximising g very likely exists, because of the first two terms on the right-hand side of Eq. (22), capturing costs in terms of subsidy given to the informal sector and increased enforcement, are negative. But, the third term on the right-hand side, capturing the favourable effect of enforcement on formal production, is positive. We assume that the optimum level of enforcement for growth g exists (denoted by g^*) when the right-hand side of Eq. (22) is zero. The second order condition, i.e., $\frac{\partial^2 v}{\partial g^2} < 0$, is satisfied (see Appendix C).

Now, to analyse properties of g^* in more details, we write below the equation by solving for g^* from Eq. (22).

$$\frac{1}{\alpha} \frac{\mu'(g)}{\mu(g)} = \frac{\bar{\tau}_G z'(g) - s \frac{\partial L^*}{\partial g}}{\left\{ 1 - \bar{\tau}_G [1 + z(g)] - s(1 - L^*) \right\}} - \frac{\beta \frac{\partial L^*}{\partial g}}{\alpha L^*}. \quad (23)$$

The left-hand side of Eq. (23) captures the benefit from rising g on the growth due to the positive externality of enforcement on the formal sector. On the other hand, a rise in g also pushes up the subsidy requirement for increased informal labours and cost for improved enforcement. Hence, it raises taxes and thereby lowers formal employment. These two effects together lower marginal productivity of private

⁸ Even if a big family assumption is not made, a similar intertemporal budget constraint can be found in the presence of homogeneous labour and perfect credit market.

capital; these are captured in the right-hand side of Eq. (23). For a lower value of g , the former must be greater than the latter. So, when the enforcement level improves, formal production rises but informal production falls. At the same time, this raises the demand for taxation to meet the increase in the number of informal workers and, thereby, affects formal production adversely. As long as the benefit of enforcement is stronger, the economy grows faster. At an extreme level of enforcement, the direct effect may be weaker than the former, and growth can slow down. Hence, an inverted-U shaped growth relationship is plausible against the level of enforcement.

It can also be shown that $\frac{\partial g^*}{\partial s} < 0$, i.e. the optimum level of enforcement for growth varies inversely with the subsidy rate (see Appendix D). The intuition behind this result is as follows. The optimum enforcement maintains a balance between the marginal cost and benefit of income from a change in the level. Now, a rise in the subsidy rate lowers formal employment and raises income tax. These two effects raise the marginal cost of enforcement at the preceding g^* , and the marginal benefit from increased formal production falls short of the marginal cost. Growth can be raised by lowering the enforcement. This negative relationship between growth and enforcement continues until a new equilibrium level for growth-maximising enforcement, corresponding to the higher subsidy rate, is reached. At the new g^* , equality is restored between marginal cost and benefit. Thus, a rise in the subsidy rate results in a fall in the optimum enforcement level for growth rate.

Similarly, it can be shown that $\frac{\partial g^*}{\partial \theta} < 0$ (see Appendix E). The intuition behind this result is the same as in the case of effecting s on g^* . A rise in θ lowers formal employment and raises income tax to subsidise the informal sector. So, the marginal cost of enforcement at the preceding g^* rises than the benefit from it. So, g^* has to fall. Elgin and Birinci (2016) find similar results empirically.

Note that when the government does not finance informal subsidy, i.e. $s = 0$, then incorporating Eqs. (13) and (19) into Eq. (22) we obtain

$$\frac{\frac{\partial v}{\partial g}}{v + \rho} = \frac{-\bar{\tau}_G z'(g)}{\{1 - \bar{\tau}_G[1 + z(g)]\}} - \left(\frac{\beta}{\alpha}\right) \frac{\left\{z'(g)\left\{\bar{\tau}_G B\gamma(g)\right\} + B\gamma'(g)\left\{1 - \bar{\tau}_G[1 + z(g)]\right\}\right)}{B\gamma(g)\left\{1 - \bar{\tau}_G[1 + z(g)]\right\}} + \frac{1}{\alpha} \frac{\mu'(g)}{\mu(g)}. \quad (22a)$$

This condition implies that the optimum enforcement level for growth (say, g^{**}) becomes independent of the relative bargaining power of the union. It can be understood by looking at the right-hand side of Eq. (22a), which is independent of θ . The optimum enforcement level is solved by equating the right-hand side of the above equation to zero. As a result, g^{**} is independent of θ . This is so because a fall in formal employment due to unionisation does not raise the tax to transfer to the informal sector in this case. Also note that a relative change in formal employment due to enforcement, i.e., $\frac{\partial \bar{L}}{\partial L}$, becomes independent of θ , because the relative bargaining power of the union enters with multiplicative form into formal employment (Eq. (13)) and the growth rate (Eq. (21)). So, the relative bargaining power of the union has no effect on g^{**} ; hence, negative enforcement due to relative bargaining power does not enter here. On the other hand, the additional pressure from increased enforcement on taxation for subsidising informal income does not arise. Hence, g^{**} can definitely be higher than g^* . This has particular relevance for public policy. If the state does not care about redistribution, the enforcement level that maximises growth rate must be higher than that with redistribution. Moreover, when informal income exceeds poverty line income, the state may not continue to subsidise informal income and hence can raise the enforcement level. The important results obtained in this subsection regarding the optimum level of enforcement are summarised in Proposition 2.

Proposition 2. *The enforcement level on growth rate shows an inverted-U*

shaped relationship. The optimum enforcement level for growth rate improves with lower subsidy rate to the informal sector and the formal labour bargaining power.

This hump-shaped relationship between growth rate and enforcement level is demonstrated in Fig. 4. The enforcement level (g) is measured on the horizontal axis, and the growth rate (v) is measured on the vertical axis. The growth curve shows an inverted-U shaped relationship against enforcement. The level corresponding to the highest point on the curve shows the optimum enforcement for growth (denoted by g^*). However, the growth curve shifts south-west due to a rise in the subsidy rate and/or labour union's bargaining power, and the optimum level of enforcement for growth falls. For example, if the subsidy rate rises from s_0 to s_1 , the growth-maximising enforcement level falls from g_1^* to g_0^* .

4.2. Enforcement and subsidy

In the earlier subsection, we investigated the relationship between the optimum enforcement for growth, g^* , and subsidy rate, s , without fixing growth rate. Often a state, specifically with a large informal sector, may be forced to raise subsidies for political and structural compulsions. Now, the trade-off between the enforcement level and subsidy rate will be investigated if the state wants to keep the growth rate constant. Totally differentiating Eq. (21) and keeping growth rate, v , as constant, we obtain

$$\left\{ \frac{-\bar{\tau}_G z'(g) + s \frac{\partial L}{\partial g}}{\{1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*)\}} + \frac{\beta}{\alpha} \frac{\frac{\partial L}{\partial g}}{L^*} + \frac{1}{\alpha} \frac{\mu'(g)}{\mu(g)} \right\} dg = \left\{ \frac{(1 - L^*) - s \frac{\partial L}{\partial s}}{\{1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*)\}} - \frac{\beta}{\alpha} \frac{\frac{\partial L}{\partial s}}{L^*} \right\} ds. \quad (24)$$

Now, Eq. (13a) implies that the terms inside the curly bracket on the right-hand side of the above equation that captures the net benefits from the rise in the subsidy rate is positive. The net benefit counts gains in informal income after subtracting the loss of formal income for the increased taxation needed to raise subsidy. However, the terms inside the curly brackets on the left-hand side that capture the net effect of enforcement change are ambiguous; the equation is $\left(\frac{\partial v}{\partial g}\right)/(v + \rho)$. The

net effect depends on the extent of formal output rise in comparison to the loss of informal income and allied costs for increased enforcement level. Its sign depends on the value of g . If $g \geq g^*$, then it is ≤ 0 . So

$\left.\frac{dg}{ds}\right|_{v=\text{constant}} \leq 0$ if $g \geq g^*$. The increased subsidy requires higher taxation from the formal sector and negatively affects income and employment. To compensate for the loss, the enforcement level needs to be manipulated. A marginal rise of enforcement could raise the net income if its favourable effect on the formal sector compensates for the cost of enforcement and loss of informal income. This occurs at a point lower than g^* and, hence, suggests that a state with a lower enforcement level must improve it to maintain growth constant when the subsidy is increased. If the state is incapable of raising the level, growth will decline. On the other hand, a state with a higher level of enforcement (i.e., $g > g^*$) needs to lower the level to maintain the growth rate for an improved subsidy. Otherwise, the growth rate will be lower. Hence, when the state undertakes higher redistributive measures, it should choose to lower their current enforcement level to maintain their growth rate.

This result can be easily illustrated with the help of a diagram (Fig. 4). Suppose the subsidy rate is increased from s_0 to s_1 . The growth curve shifts south-west from $v(s_0)$ to $v(s_1)$. The growth-maximising level of enforcements are g_0^* and g_1^* respectively. Suppose that the government could not choose the growth-maximising level of enforcement and, the economy was growing at the rate $\bar{v}$ with subsidy rate s_0 . After a hike in subsidy rate to s_1 , the economy wants to grow at the same rate, $\bar{v}$. We can

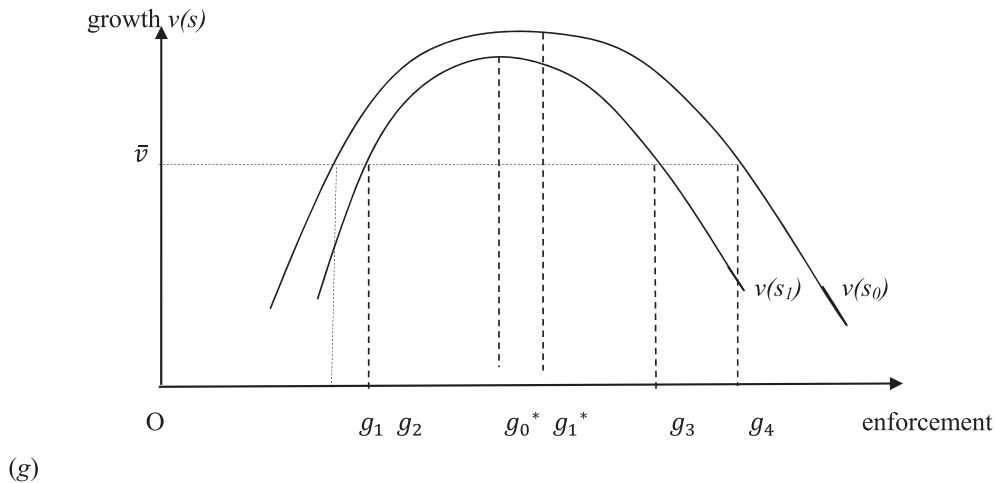


Fig. 4. Enforcement and subsidy with a constant economic growth rate.

see two cases. In the first case, when $g_1 < g_1^*$, the enforcement has to rise to g_2 to maintain the same growth rate. In the second case, when $g_4 < g_1^*$, the enforcement has to fall to g_3 to maintain the same growth rate. This result is stated in [Proposition 3](#).

Proposition 3. *If a state raises the subsidy rate and does not want to sacrifice growth rate, a state with a higher level of enforcement than the optimum should decrease the level; a state with a lower level of enforcement than the optimum should raise the level, and otherwise accept lower growth.*

In reality, typical developing countries like India, Brazil and Bangladesh spend a lower subsidy, where lower enforcement serves as social security for the informal sector. On the other hand, we also see that developed countries, especially in Europe, incur higher subsidy along with higher enforcement. The result seems to be suggesting a positive association between enforcement level and subsidy rate for countries with relatively weaker enforcement if they want to improve redistribution without sacrificing growth. However, our result obtained in [subsection 4.1](#) shows that optimum enforcement varies inversely with the subsidy rate. The incumbent parties in a democratic society may not be keen to maximise the growth rate ([Choi and Thum, 2005](#)); rather, they may prefer to raise formal employment by giving the formal sector tax concessions. Then, the enforcement level must be lowered to encourage informal income to some extent for their incumbency. Here, Eqs. (19) and (20) satisfy these two objectives simultaneously together with the presence of informal employment. According to [Proposition 3](#), if a state with weaker enforcement raises subsidy for the informal sector, it needs to improve enforcement without sacrificing the growth rate. Then, the subsidy rate and the enforcement level move in the same direction.

4.3. Optimum enforcement for welfare

In this subsection, the optimum levels of enforcement that maximise growth and welfare are compared. Welfare, ω , is defined as the discounted present value of instantaneous utility of the representative household, as follows:

$$\omega = \int_0^\infty \log(c) e^{-\rho t} dt \quad (25)$$

Now $c = c_0 e^{vt}$, where, c_0 denotes consumption at the beginning. After incorporating the expression of c into Eq. (25) and integrating that, we obtain

$$\omega = \frac{\log(c_0)}{\rho} + v \left[\frac{1}{\rho^2} + x_1 \right] + x_2 \quad (25a)$$

where, x_1 and x_2 are constants of integration. To obtain c_0 , we use Eqs. (1)–(3), (6)–(8) and (17) and (21) and get

$$c_0 = K_0 \left\{ \frac{(v + \rho) [s(1 - L^*) + \bar{\tau}_G B \gamma(g)(1 - L^*)]}{\alpha [1 - \bar{\tau}_G [1 + z(g)] - s(1 - L^*)]} + \frac{[v(1 - \alpha) + \rho]}{\alpha} \right\}. \quad (26)$$

Eqs. (25a) and (26) show that if ω is differentiated with respect to g at $g = g^*$, then it may not be zero.

$$\frac{\partial \omega}{\partial g} \Big|_{g=g^*} = \frac{1}{\rho c_0} \frac{\partial c_0}{\partial g} \Big|_{g=g^*} \quad (27)$$

where,

$$\begin{aligned} \frac{\partial c_0}{\partial g} \Big|_{g=g^*} &= \left\{ \frac{K_0(v + \rho)}{\alpha [1 - \bar{\tau}_G [1 + z(g)] - s(1 - L^*)]^2} \right\} \Big|_{g=g^*} \\ \frac{\partial c_0}{\partial g} \Big|_{g=g^*} &= \left\{ \frac{K_0(v + \rho)}{\alpha [1 - \bar{\tau}_G [1 + z(g)] - s(1 - L^*)]^2} \right\} \Big|_{g=g^*} \\ &\quad \left\{ \left\{ \bar{\tau}_G B \gamma'(g)(1 - L^*) - [\bar{\tau}_G B \gamma(g) + s] \frac{\partial L^*}{\partial g} \right\} [1 - \bar{\tau}_G [1 + z(g)] - s(1 - L^*)] \right. \\ &\quad \left. + [s(1 - L^*) + \bar{\tau}_G B \gamma(g)(1 - L^*)] \left\{ \bar{\tau}_G z'(g) - s \frac{\partial L^*}{\partial g} \right\} \right\} \Big|_{g=g^*} \end{aligned} \quad (28)$$

$$\text{So, } \frac{\partial \omega}{\partial g} \Big|_{g=g^*} \gtrless 0 \text{ as } \frac{\partial c_0}{\partial g} \Big|_{g=g^*} \gtrless 0.$$

This is because welfare depends on two things – the initial level of consumption and the growth rate. In this model, the growth rate is positively related to the net marginal productivity of capital. But, the enforcement level affects the marginal productivity of capital through three channels – taxation, formal employment and a direct positive externality to the formal sector. When the enforcement level affects informal income, it influences initial consumption, but it does not affect the marginal productivity of private capital after tax. Hence, it cannot affect the growth rate through this channel. As a result, the effect of the enforcement level on initial consumption may be different from the one that maximises the growth rate. To be more specific, by looking at Eq. (26), it becomes clear that in the absence of informal income and subsidy, i.e., when $B = s = 0$, initial consumption is maximised at the enforcement level that maximises the growth rate. However, when households have other than formal income, an increase in enforcement level has a negative and direct impact on the marginal income from the informal sector.

This is captured by the first term inside the second bracket on the right-hand side of Eq. (28). On the other hand, it raises informal employment and, hence, needs taxation on formal income to subsidise them. Also, increased taxation reduces formal output. These two effects are captured by the second term inside the second bracket on the right-hand side of Eq. (28). Hence, the initial level of consumption should not be maximised at the growth rate–maximising the level of g .

Mathematically, the first term in the right-hand side of Eq. (28), $\left\{ \frac{K_0(v+\rho)}{\alpha[1-\bar{\tau}_G[1+z(g)]-s(1-L^*)]^2} \right\} \Big|_{g=g^*}$, is positive. As a result, the algebraic sign of the right-hand side of Eq. (28) will solely depend on the second term. Since only $\bar{\tau}_G B\gamma'(g)(1-L^*)[1-\bar{\tau}_G[1+z(g)]-s(1-L^*)]$ term is negative in the term $\frac{\partial c_0}{\partial g} \Big|_{g=g^*}$ expression (see Eq. (19) and Condition 2 in Appendix B), this negative part will be dominated by all other positive terms, and hence $\frac{\partial \omega}{\partial g} \Big|_{g=g^*}$ becomes positive.

This means that the optimum enforcement level for welfare is greater than the level that maximises the growth rate. This conjecture is confirmed in the simulation exercise undertaken in subsection 4.5. Note that when the government does not subsidise informal income (i.e. $s = 0$), we obtain using Eqs. (13) and (26),

$$c_0 = K_0 \left\{ \frac{(v+\rho)\bar{\tau}_G B\gamma(g)}{\alpha[1-\bar{\tau}_G[1+z(g)]]} - \frac{(v+\rho)\Theta}{\alpha} + \frac{[v(1-\alpha)+\rho]}{\alpha} \right\}. \quad (26a)$$

This implies that unlike enforcement-maximising growth rate, the optimum enforcement for welfare is not independent of a union's bargaining power. The increased union power does not affect the rate of capital formation from formal income in the absence of subsidy, but it can change welfare because initial consumption is affected by the enforcement level. Since increased enforcement hurts informal income and, thereby, on the initial consumption of the household, the level that maximises welfare without subsidy must be even higher so that the increased formal income can compensate the loss.

Proposition 4. *The optimum enforcement for welfare is likely to be greater than the one maximising growth rate due to its effect on the initial consumption. The level depends on the union bargaining power of formal labour even when the informal sector is not subsidised.*

An Extension: Note that household utility is assumed to depend solely on consumption in the previous exercise. In reality, better enforcement could be another external source of household utility. In that case, the utility function would look like

$$\int_0^\infty [\log(c)\eta(g)]e^{-\rho t} dt; \eta(g) > 1, \eta'(g) > 0 \text{ and } \eta''(g) < 0. \quad (16a)$$

where, $\eta(g)$ captures the impact of enforcement on household utility.

The marginal impact of enforcement is positive but diminishing. We also assume that the value of $\eta(\cdot)$ function at the lowest possible level of enforcement is greater than unity. Introducing this additional phenomenon will not change our results qualitatively. However, this is an additional positive effect of enforcement on welfare. As a result, the optimum level of enforcement for welfare will be higher than what we get in the previous case. The mathematical counterpart of the welfare function given by Eq. (25a) will be

$$\omega = \frac{\log(c_0)\eta(g)}{\rho} + \eta(g)v\left[\frac{1}{\rho^2} + x_1\right] + x_2. \quad (25b)$$

The value of initial consumption level, c_0 , is similar to that given by Eq. (26). As a result, the differentiation of welfare with respect to enforcement at the growth-maximising level is given by

$$\frac{\partial \omega}{\partial g} \Big|_{g=g^*} = \frac{\eta(g)}{\rho c_0} \frac{\partial c_0}{\partial g} \Big|_{g=g^*} + \frac{\log(c_0)\eta'(g)}{\rho} \Big|_{g=g^*} + \eta'(g)v\left[\frac{1}{\rho^2} + x_1\right] \Big|_{g=g^*}. \quad (27a)$$

As is easily understood by comparing Eqs. (27) and (27a), the optimum enforcement level for welfare will be higher than the one obtained in the absence of its inclusion in household utility. This result is stated below as a corollary.

Corollary 1. *If better enforcement has a positive external effect on household utility, the gap between optimum levels of enforcement respectively for growth rate and welfare widens.*

4.4. Variable subsidy rate

For simplicity, in the earlier analysis, the subsidy per informal worker was assumed to be an exogenously given fraction of formal output. However, it may be possible that the government provides social security or subsidy only to the poor (i.e. informal workers in this model) until it ensures a certain standard of living (say, poverty line income). In that case, if the government fixes the level of subsidy per worker, the fraction of formal output used as subsidy per worker will continuously decrease with formal sector growth. In that case, the growth rate and optimum level of enforcement will be time-dependent. This can be shown easily. From Eq. (21), given the enforcement level, an increase in subsidy rate lowers the growth rate, because this lowers formal employment but raises taxation.

$$\frac{\frac{\partial v}{\partial s}}{v+\rho} = \frac{-(1-L^*) + s \frac{\partial L^*}{\partial s}}{\left\{1 - \bar{\tau}_G[1+z(g)] - s(1-L^*)\right\}} + \frac{\beta}{\alpha} \frac{\frac{\partial L^*}{\partial s}}{L^*} < 0. \quad (29)$$

As time passes, the subsidy rate, s , gets smaller in size and the growth curve shifts upward in the enforcement–growth plane. We have already shown that the optimum level of enforcement for growth increases with a fall in subsidy rate, s . So, the growth curve shifts north-east (Fig. 5).

Suppose that there are two different subsidy rates s_0 and s_1 , where $s_0 > s_1$. $v(s_0)$ is the growth curve corresponding to the subsidy rate s_0 , and $v(s_1)$ is the growth curve corresponding to the subsidy rate s_1 . The optimum levels of enforcement for growth rate are g_0^* and g_1^* , respectively, where $g_0^* < g_1^*$. When the government endogenously determines the subsidy rate to maintain certain informal income (or fixed subsidy per worker, in our case), the rate gradually declines with the expansion of the formal sector, growth rate and optimum level of enforcement over time. This is summarised in Proposition 5.

Proposition 5. *If a state endogenously determines subsidy rate to keep informal income constant over time, the subsidy rate gradually falls with the expansion of the formal sector, growth rate and enforcement level.*

4.5. Simulation

The subsection attempts to simulate some of the key relationships discussed in the previous subsection. This requires us to find standard

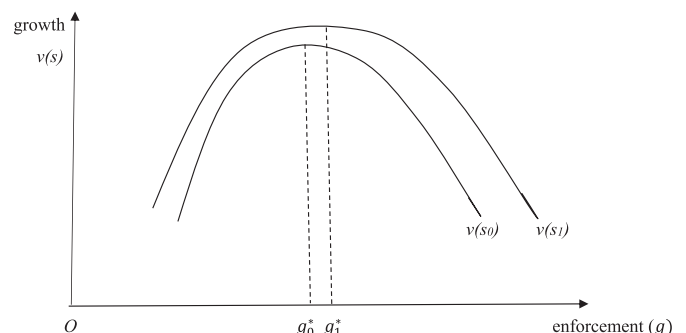


Fig. 5. Variable subsidy rate, enforcement level and growth.

Table 2
Parameters used for simulation.

Parameters	Values	Sources/assumptions
capital share(α)	0.30	Assumed such that $\alpha + \beta < 1$
labour share (β)	0.60	Douglas (1976)
A	1	By choice of unit
B	0.20	Marjit and Maiti (2009)
The rate of discount factor (ρ)	0.02	Barro and Sala-i-Martin (2004)
$\gamma = 50 - g^2$		Assumed in Jonasson (2011)
$z = 0.2g^2$		Assumed in Jonasson (2011)
$\theta = 0.3$		Abraham et al. (2009)
M	0.6	Gahan (2002)
N	0.4	Gahan (2002)
K0	1	By choice of units
$\bar{\tau}_G$	0.10	Based on RBI statistics
S	0.05	Based on RBI statistics

values of the parameters relevant for a typical economy seen in the literature (see Table 2). In the production function of the formal sector, we assume that capital share (α) = 0.30 and labour share (β) = 0.60 with the level of technology (A) = 1. Enforcement externality is specified as $\mu = 3g^{0.5}$. We assume the elasticity of enforcement effectiveness here is 0.5.

It has been observed that the level of productivity in the formal sector is roughly three to five times as high as that of the informal sector on an average (Marjit and Maiti, 2009). Hence, B is assumed to be 0.2. Since informal activities depend highly on the level of enforcement, we assume that $\gamma = 50 - g^2$. The intercept term is kept at a higher level so that the marginal effect of enforcement remains negative, but γ becomes positive. This specification also suggests that the marginal effect of informal production falls below 0.4 (as shown in Jonasson, 2011). Hence, the cost of enforcement is defined as $z = 0.2g^2$. The cost of bureaucratic resources is found 0.42 in Jonasson (2011).

On average, 10% of formal sector output is taxed in a typical developing economy and hence $\bar{\tau}_G$ is assumed to be 0.10. According to the Reserve Bank of India, the economy roughly spends 5% of GDP to the social welfare schemes and subsidies. The subsidy, therefore, for social welfare scheme is considered to 5% of GDP ($s = 0.05$).

Usually, a union assigns a higher weight on wage than employment (see Gahan, 2002, Table 3). Hence, we assume that $m = 0.6$ and $n = 0.4$. Abraham et al. (2009) estimated that union bargaining power is roughly 0.2 to 0.3 in firms in Belgium. Using a similar method, Maiti (2014) suggested that it varies from 0.1 to 0.5 across formal sector firms in India. Following these studies, we take an average and specify $\theta = 0.3$. The initial level of capital stock (K_0) is assumed as one because its scale does not have much implication on our calibrated values. The rate of discount factor (ρ) is considered as 0.02 (Barro and Sala-i-Martin, 2004).

Given the specification mentioned above, we have calibrated the

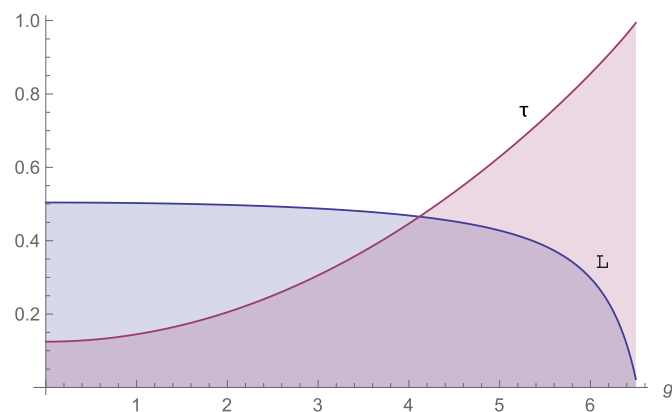


Fig. 6. Formal employment (L) and taxation (τ) against enforcement level (g) in equilibrium.

values of formal employment, taxation and growth rate. First, we find that the enforcement level that satisfies our conditions varies from roughly 0 to 6.5. Any value higher than 6.5 does not satisfy those conditions. Within this range, formal employment (L^*), derived from Eq. (13), registers an inverse relationship with the enforcement level (Fig. 6). In other words, formal employment declines with the rise of enforcement, and this satisfies our conjecture.

On the other hand, we derive taxation (τ) from Eq. (8). The taxation rises when enforcement improves to compensate informal income (Fig. 6). This also confirms our hypothesis. Further, the growth rate is estimated using Eq. (21) (Fig. 3) and welfare is estimated using Eq. (25) (Fig. 4). Both the demonstrated curves are inverted-U shaped against the enforcement level. Interestingly, the optimum level of enforcement for welfare is found at $g = 4.4$. On the other hand, the optimum enforcement level for growth rate is observed at $g = 4.2$ (Fig. 7). So, the optimum enforcement level for the growth rate is lower than that for welfare. More importantly, both of them fall in between the two extreme levels of enforcement. This confirms that in a typically developed economy, deliberately weak enforcement improves both growth rate and welfare.

4.6. An extension with competitive labour market

In subsections 4.3 and 4.4, we formulate the theoretical model with a unionised labour market. However, labour unions are weaker in developing countries than in advanced countries, and their power varies considerably; so, it becomes necessary to validate the results obtained from our model in a setting without unions.

In the absence of a union, the market determines the allocation of labour between the two sectors by equating their marginal productivities. Since, due to technological differences, the marginal product of formal workers is higher than that of informal workers, the strategic redistributive policy is still very much relevant. The results of this analysis can be obtained by putting $\theta = 0$ in the previous analysis. In the whole analysis, the union's effect is captured by the term θ . Before going further, we put

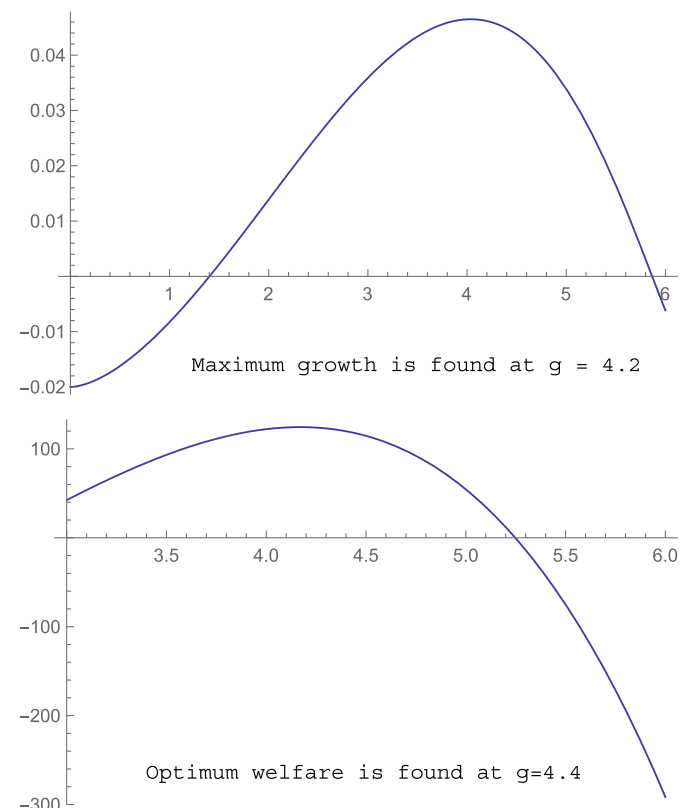


Fig. 7. Optimum levels of enforcement for growth and welfare.

$\theta = 0$ in Eq. (14) and obtain $\Theta = \beta$. Substituting β in the place of θ in the previous theoretical analysis, it gives us the results for the non-unionised case. The theoretical results derived in the case of unionised labour market remain qualitatively similar to the case with non-unionised labour markets. Since the absence of a union would increase the favourable effect of enforcement on the formal sector and raise employment, the growth curve shifts upward, and the optimum level of enforcement must rise to a higher level than the one with the union.

5. Conclusions

This paper shows that the enforcement level, which encompasses the level of security of property rights, the integrity of contracts and checks of corruptions, can deliberately be chosen by a state with informality to limit the burden on tax-payers for redistribution. While the enforcement choice affects formal and informal activities differently and taxation reduces formal activities only, their simultaneous choices imply the growth as well. We develop a two-sector growth of an economy with informality, where workers who do not find employment in the formal sector find something in the informal sector for survival. The duality arises due to the technological difference between sectors and union in the formal sector. The taxes collected from the formal sector is spent on public infrastructure that facilitates production in both sectors, and on subsidies to the informal sector. If the enforcement favours the formal sector and discourages the informal activities, the state can endogenously choose the enforcement level along with taxation.

In such a framework, we find that taxation and enforcement are positively related. A relatively lower level of enforcement allows the informal sector to survive to some extent and, hence, reduces the distortionary effect of formal sector production from higher taxation. This raises growth. If the enforcement level is raised, a state that cares about redistribution must raise taxes to subsidise the informal sector, which

will limit growth. Moreover, if enforcement is too weak, the formal sector would be severely affected, and growth could be dampened. We find an inverted-U shaped growth curve against the enforcement level. There exists a unique level of enforcement that maximises the growth rate. However, the optimum enforcement level rises with a lower subsidy for informal sector workers and weaker bargaining power of formal sector labour. When the state does not subsidise informal income, the optimum enforcement level for growth must be higher than the one with a subsidy. The optimum level of enforcement for welfare is a bit higher than the one for growth, specifically when consumers care about the quality of enforcement. Considering standard parametric values, we simulated the model and found that taxation is positively related to the enforcement level. The growth rate and welfare functions are found inverted-U shaped against the enforcement level. Both the optimum levels for growth rate and welfare are found at lower than the extreme values. However, the optimum enforcement level for welfare would be higher than the one maximising growth rate.

Hence, we argue that a benevolent state, which cares about distribution, prefers to choose the enforcement level and taxation such that growth is maximised along with better redistribution. If the state ignores redistribution totally, the growth rate could be raised at the cost of informal sector welfare. When the informal sector is sufficiently large, this would likely raise social unrest, and the state may face a threat to its incumbency. Hence, the state cannot ignore the welfare of the informal sector for its incumbency. Moreover, when both inequality and informality are thriving fast across the world, and fiscal deficits are targeted to limit after crises during the past decade, the state should not raise the enforcement level strictly unless taxation can be raised. Hence, a democratic government with an informal sector prefers to weaken the enforcement level a bit to keep taxation low and ensure its incumbency. This improves both growth rate and welfare as well.

Appendix A

To have Θ positive, we need the following condition to be satisfied.

Condition 1: $\theta n(1 - \alpha - \beta) + \beta(1 - \theta)(1 - \beta) > \theta m(1 - \beta)(1 - \alpha - \beta)$

This is satisfied trivially when $m \leq n$, i.e., the union is ‘employment oriented’ or ‘neutral’. Even if the union is wage-oriented, this condition still holds, provided that the union is not excessively wage-oriented.

Appendix B

Both numerator and denominator of the right-hand side of Eq. (13) have to be positive, and the denominator has to be greater than the numerator. They are satisfied under the following conditions 2, 3 and 4.

Condition 2: $1 - \bar{\tau}_G[1 + z(g)] - s > 0$;

Condition 3: $\bar{\tau}_G B_Y(g) > s\Theta$;

Condition 4: $\Theta - \Theta \bar{\tau}_G[1 + z(g)] < \bar{\tau}_G B_Y(g)$.

Conditions (2), (3) and (4) impose an upper bound on the level of governance for given $\bar{\tau}_G$ and s .

Appendix C

For the second-order condition of maximisation, we use Eq. (22) and obtain

$$\frac{\frac{\partial^2 v}{\partial g^2}}{v + \rho} = \frac{\left[-\bar{\tau}_G z''(g) + s \frac{\partial^2 L^*}{\partial g^2} \right] \left\{ 1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*) \right\} - \left[-\bar{\tau}_G z''(g) + s \frac{\partial^2 L^*}{\partial g^2} \right]^2}{\left\{ 1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*) \right\}^2} + \frac{\beta \frac{\partial^2 L^*}{\partial g^2} L^* - \left(\frac{\partial L^*}{\partial g} \right)^2}{(L^*)^2} + \frac{1}{\alpha} \frac{\mu''(g)\mu(g) - [\mu'(g)]^2}{[\mu(g)]^2}. \quad (C1)$$

We further assume $\frac{\partial^2 L^*}{\partial g^2} < 0$, which is a sufficient, but not a necessary condition for the right-hand side of this equation to be negative.

Appendix D

To analyse the effect of a rise in the subsidy rate, s , on g^* , we totally differentiate Eq. (23), and obtain

$$\left\{ \frac{1}{\alpha} \frac{\mu''(g)\mu(g) - [\mu'(g)]^2}{[\mu(g)]^2} + \frac{\beta}{\alpha} \frac{\frac{\partial^2 L^*}{\partial g^2} L^* - \left(\frac{\partial L^*}{\partial g}\right)^2}{(L^*)^2} + \frac{[-\bar{\tau}_G z''(g) + s \frac{\partial^2 L^*}{\partial g^2}]}{\left\{1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*)\right\}^2} \left\{1 - \bar{\tau}_G z''(g) + s \frac{\partial^2 L^*}{\partial g^2}\right\} \right\} dg$$

$$= \left\{ \frac{\left\{1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*)\right\} \left\{-\frac{\partial L^*}{\partial g} - s \frac{\partial^2 L^*}{\partial s \partial g}\right\} - \left\{\bar{\tau}_G z'(g) - s \frac{\partial L^*}{\partial g}\right\} \left\{-(1 - L^*) + s \frac{\partial L^*}{\partial s}\right\}}{\left\{1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*)\right\}^2} - \frac{\beta}{\alpha} \frac{\frac{\partial^2 L^*}{\partial s \partial g} L^* - \frac{\partial L^*}{\partial g} \frac{\partial L^*}{\partial s}}{(L^*)^2} \right\} ds. \quad (D1)$$

The term inside the big curly bracket on the left-hand side of this equation is negative as $\frac{\partial^2 v}{\partial g^2} < 0$ (See Appendix C). To get a sign of the right-hand side of this equation, we use Eq. (19) and obtain (using conditions 4 and 5):

$$\frac{\partial^2 L^*}{\partial s \partial g} = - \frac{\bar{\tau}_G \Theta \left[\Theta \left\{ z'(g) \left\{ \bar{\tau}_G B\gamma(g) - s\Theta \right\} + B\gamma'(g) \left\{ 1 - \bar{\tau}_G[1 + z(g)] - s \right\} \right\} - B\gamma'(g) \left[\bar{\tau}_G B\gamma(g) - \Theta \left\{ 1 - \bar{\tau}_G[1 + z(g)] \right\} \right] \right]}{\left\{ \bar{\tau}_G B\gamma(g) - s\Theta \right\}^3} < 0. \quad (D2)$$

Equations (13.a), (19) and (D.2) imply $\frac{\partial^2 L^*}{\partial s \partial g} L^* - \frac{\partial L^*}{\partial g} \frac{\partial L^*}{\partial s} < 0$. On the other hand, these equations also imply the other part of the right-hand side of Eq. (D.1) to be positive. Hence, $\frac{\partial g^*}{\partial s} < 0$.

Appendix E

We derive the effect of a rise in the union's bargaining power on g^* by totally differentiating Eq. (23).

$$\left\{ \frac{1}{\alpha} \frac{\mu''(g)\mu(g) - [\mu'(g)]^2}{[\mu(g)]^2} + \frac{\beta}{\alpha} \frac{\frac{\partial^2 L^*}{\partial g^2} L^* - \left(\frac{\partial L^*}{\partial g}\right)^2}{(L^*)^2} + \frac{[-\bar{\tau}_G z''(g) + s \frac{\partial^2 L^*}{\partial g^2}]}{\left\{1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*)\right\}^2} \left\{1 - \bar{\tau}_G z''(g) + s \frac{\partial^2 L^*}{\partial g^2}\right\} \right\} dg$$

$$= \left\{ \frac{\left\{1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*)\right\} \left\{-s \frac{\partial^2 L^*}{\partial \theta \partial g}\right\} - \left\{\bar{\tau}_G z'(g) - s \frac{\partial L^*}{\partial g}\right\} \left\{s \frac{\partial L^*}{\partial \theta}\right\}}{\left\{1 - \bar{\tau}_G[1 + z(g)] - s(1 - L^*)\right\}^2} - \frac{\beta}{\alpha} \frac{\frac{\partial^2 L^*}{\partial \theta \partial g} L^* - \frac{\partial L^*}{\partial g} \frac{\partial L^*}{\partial \theta}}{(L^*)^2} \right\} d\theta. \quad (E1)$$

The term inside the big curly bracket on the left-hand side of Eq. (E.1) is negative as $\frac{\partial^2 v}{\partial g^2} < 0$ (See Appendix C). To sign the right-hand side of Eq. (E.1), we use Eq. (19) and obtain:

$$\frac{\partial^2 L^*}{\partial \theta \partial g} = - \frac{\bar{\tau}_G \frac{\partial \Theta}{\partial \theta} \left[\bar{\tau}_G B\gamma(g) z'(g) \left\{ \bar{\tau}_G B\gamma(g) - s\Theta \right\} + B\gamma'(g) \left\{ 1 - \bar{\tau}_G[1 + z(g)] - s \right\} \left\{ \bar{\tau}_G B\gamma(g) + s\Theta \right\} \right]}{\left\{ \bar{\tau}_G B\gamma(g) - s\Theta \right\}^3}. \quad (E2)$$

We assume the following condition to hold.

Condition 6: $z'(g) \left\{ \bar{\tau}_G B\gamma(g) - s\Theta \right\} + B\gamma'(g) \left\{ 1 - \bar{\tau}_G[1 + z(g)] - s \right\} \frac{\left\{ \bar{\tau}_G B\gamma(g) + s\Theta \right\}}{\bar{\tau}_G B\gamma(g)} < 0$.

This condition implies $\frac{\partial^2 L^*}{\partial \theta \partial g} < 0$. So, equations (13.c), (19) and (E.2) imply $\frac{\partial^2 L^*}{\partial \theta \partial g} L^* - \frac{\partial L^*}{\partial g} \frac{\partial L^*}{\partial \theta} < 0$. On the other hand, this set of equations also imply the other part of the right-hand side of Eq. (E1) to be positive. So $\frac{\partial g^*}{\partial \theta} < 0$.

Appendix F. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.econmod.2019.04.015>.

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